Inputs

Project Description

This project uses three LEDs to create a traffic light.

⋮

When the Arduino powers up, it should:
1. Configure the button pin as INPUT.
2. Configure three LED pins as OUTPUT.
3. Continuously monitor the button state.
4. If it is not pressed, turn the green LED3 on.
5. If it is pressed, turn the green LED3 off, and turn the yellow LED2 on for 1000ms, and then turn the red LED1 on for 2000ms. Lastly, turn the green LED3 on.

Code

```
void setup() {···}
void loop() {
  if (digitalRead(BUTTON_PIN)==HIGH) {
    digitalWrite(GREEN_LED, LOW);
    digitalWrite(YELLOW_LED, HIGH);
    delay(1000);
    ···
  }
  digitalWrite(GREEN_LED, HIGH);
  delay(500);
}
```

Evaluation

Test Procedure

Check if green LED is on

Press button
→ Check if yellow LED is on

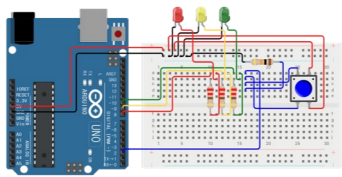
Wait 1 sec
→ Check if red LED is on

Wait 2 sec
→ Check if green LED is on

Success or **Failure**

Generated Circuit

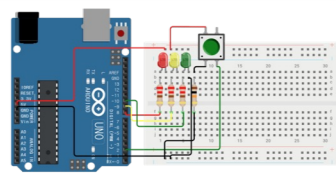
o3-Mini



✅ **Success**

Redundant Connection	0
Extraneous Component	0
Missing Component	0
Isolated Component	0
Pin Conflict	0
Breadboard Bypass	0

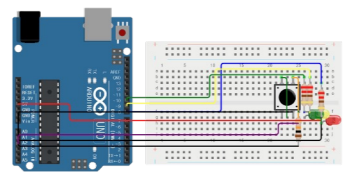
Gemini-2.0-Flash



❌ **Failure**

Redundant Connection	0
Extraneous Component	0
Missing Component	0
Isolated Component	0
Pin Conflict	**1**
Breadboard Bypass	**4**

Mistral-Small 3

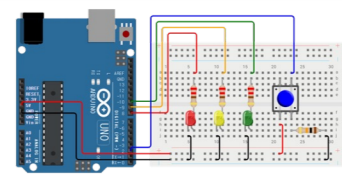


❌ **Failure**

Redundant Connection	0
Extraneous Component	0
Missing Component	0
Isolated Component	0
Pin Conflict	**4**
Breadboard Bypass	**4**

Physical Circuit

Reference



✅ **Success**

Redundant Connection	0
Extraneous Component	0
Missing Component	0
Isolated Component	0
Pin Conflict	0
Breadboard Bypass	0